

Deep Space Optical Communications (DSOC) 1550 nm Laser Uplink

1. Laser Transmitter Architecture

The optical communications terminal utilizes a 5-watt erbium-doped fiber laser operating at 1550 nanometers wavelength, coupled to a 22-centimeter aperture telescope.

2. Ground Station Acquisition

Uplink beacons from Earth Palomar 5-meter telescope are acquired using photon-counting superconducting nanowire detector arrays, achieving 100 Mbps data bandwidth across 2.2 AU.

3. Pointing & Jitter Isolation

Piezoelectric fast-steering mirrors compensate for spacecraft micro-vibrations with 0.8 microradian pointing accuracy, counteracting Martian planetary motion and orbit drift.